WORKED SOLUTION

STEP II 2011, Question 9

Mechanics • Collisions and Restitution

STEP II 2011, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

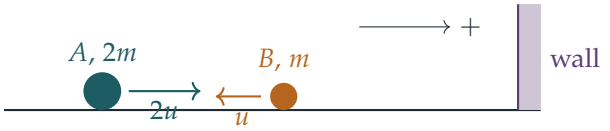
step.maths.org/questions/11-s2-q9

The key idea

Every collision here is direct, so each one is settled by the same two equations: conservation of momentum and Newton's law of restitution. The only real danger is signs. Fix one positive direction (towards the wall) at the start, write every velocity as a signed number in that direction, and the algebra takes care of itself. The final claim then comes down to showing that one expression stays positive, and since it is linear in f we only need to look at the ends of the range of f.

Solution

The model. We treat A and B as particles on a smooth plane, so no horizontal forces act except at impacts, and the wall is smooth, so it only reverses B's velocity (scaled by f). After the first collision B goes on to hit the wall, so the wall lies on B's side, beyond B in the direction A was originally moving. We take that direction as positive. The order along the line is then always A, B, wall.



The first collision

Before impact A has velocity $2u$ and B has velocity $-u$. Let their velocities afterwards be v_A and v_B. Momentum is conserved:

$$2m(2u) + m(-u) = 2m\,v_A + m\,v_B \quad \Longrightarrow \quad 2v_A + v_B = 3u.$$

The speed of approach is $2u - (-u) = 3u$. Since B is in front of A, the speed of separation is $v_B - v_A$, and Newton's law of restitution gives

$$v_B - v_A = 3eu.$$

Subtracting the second equation from the first gives $3v_A = 3u - 3eu$, so

$$v_A = (1-e)u, \qquad v_B = (1+2e)u.$$

Both are ≥ 0, so after the collision A moves towards the wall with speed $(1-e)u$ (and is at rest if $e = 1$), and B moves towards the wall with speed $(1+2e)u$. This is consistent with B then reaching the wall.

The wall

The wall is smooth and the impact is direct, so B rebounds along the line with speed $f(1+2e)u$. Its velocity becomes $-f(1+2e)u$. Meanwhile A is still moving with velocity $(1-e)u \geq 0$, because nothing acts on it. B is now moving towards A, so the two will certainly meet again.

The second collision

Just before the second impact the velocities are

$$A: \ (1-e)u, \qquad B: \ -f(1+2e)u.$$

Let the velocities afterwards be w_A and w_B. Momentum gives

$$2w_A + w_B = 2(1-e)u - f(1+2e)u.$$

Ask yourself Which way round does the restitution equation go this time?

B is still the particle nearer the wall, so the separation velocity is still $w_B - w_A$. The speed of approach is A's velocity minus B's, namely $(1-e)u + f(1+2e)u$. Hence

$$w_B - w_A = e[(1-e)u + f(1+2e)u].$$

To isolate w_B, add twice this equation to the momentum equation:

$$\begin{aligned} 3w_B &= 2(1-e)u - f(1+2e)u + 2e(1-e)u + 2ef(1+2e)u \\ &= 2(1-e)(1+e)u + f(1+2e)(2e-1)u \\ &= 2(1-e^2)u - (1-4e^2)fu. \end{aligned}$$

Therefore

$$w_B = \tfrac{2}{3}(1-e^2)u - \tfrac{1}{3}(1-4e^2)fu,$$

measured in the positive direction, that is, towards the wall. This is the required result.

B always moves towards the wall

We must show that $w_B > 0$ for all $0 < e \leq 1$ and $0 < f \leq 1$. Since $u > 0$, this is the same as showing

$$g(f) = 2(1-e^2) + (4e^2-1)f > 0.$$

The term $1 - 4e^2$ can have either sign, so we can't simply drop it. Instead, for fixed e, note that g is linear in f, and

$$g(0) = 2(1 - e^2) \geq 0, \qquad g(1) = 2 - 2e^2 + 4e^2 - 1 = 1 + 2e^2 > 0.$$

A linear function on $[0, 1]$ is the weighted average of its end values: $g(f) = (1 - f)\, g(0) + f\, g(1)$. For $0 < f \leq 1$ the first term is ≥ 0 and the second is strictly positive (as $f > 0$ and $g(1) > 0$), so $g(f) > 0$. Hence $w_B > 0$, and B moves towards the wall after the second collision, whatever the values of e and f.

Checks

- With $e = f = 1$ (perfectly elastic throughout), A stops dead at the first impact, B goes out at $3u$, comes back at $3u$, and hits a stationary particle of twice its mass. The formula gives $w_B = 0 + u = u$, which matches the elastic result $\frac{m-2m}{3m}(-3u) = u$.
- With $e = f = \frac{1}{2}$, the formula gives $w_B = \frac{2}{3} \cdot \frac{3}{4} u = \frac{1}{2} u$. A step-by-step simulation of positions and impacts gives the same value, $0.5u$.
- The worst case for the sign is $e = 1$ with f small. Then $w_B = fu$, which is small but still positive.

Common slips

- Using speeds instead of signed velocities in the second collision, and so getting B's rebound velocity with the wrong sign.
- Writing the restitution equation as $w_A - w_B$ because B "came from the wall". The order of the particles along the line hasn't changed, so separation is still $w_B - w_A$.
- Claiming $w_B > 0$ by assuming $1 - 4e^2 > 0$. It fails for $e > \frac{1}{2}$, which is why the argument has to use $f \leq 1$.

Beyond STEP

Restitution is an empirical model. In a first-year course you meet it alongside the cleaner idea of the centre-of-mass frame: in that frame the total momentum is zero, and a collision with coefficient e simply multiplies both velocities by $-e$. This makes results like the first collision almost immediate. Repeated collisions with a wall turn up again in billiard problems and in counting arguments, such as the famous one where the number of collisions between two blocks and a wall gives the digits of π. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum and collisions in this spirit.

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